

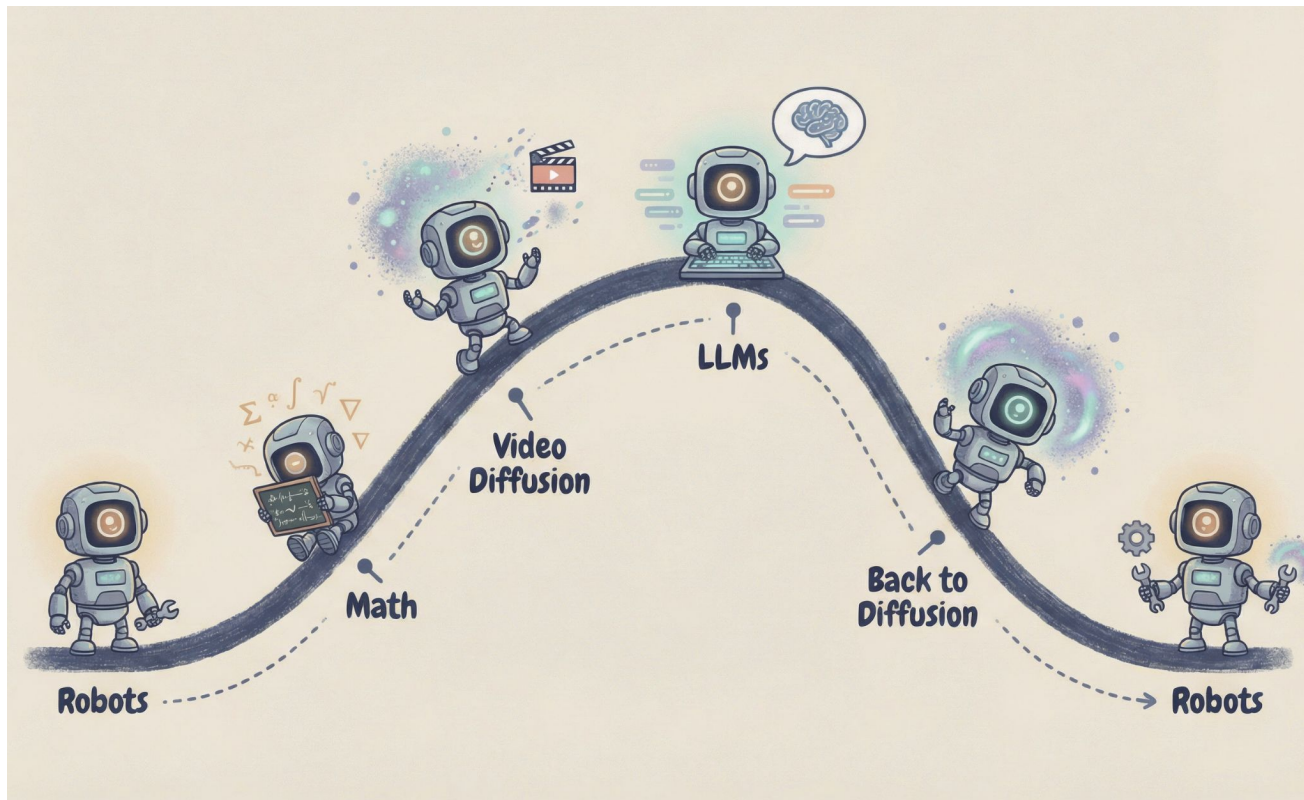
Scaling Laws for Diffusion Models

Kyle Chickering - Luma AI

Outline

- What are scaling laws?
- Why should I care?
- Our work on scaling diffusion
- What makes diffusion different?
- Robotics & Scaling - Bitter lessons?

About Me

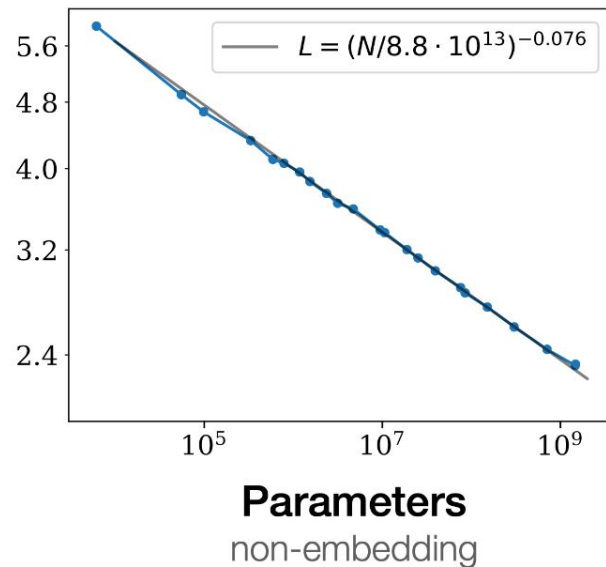
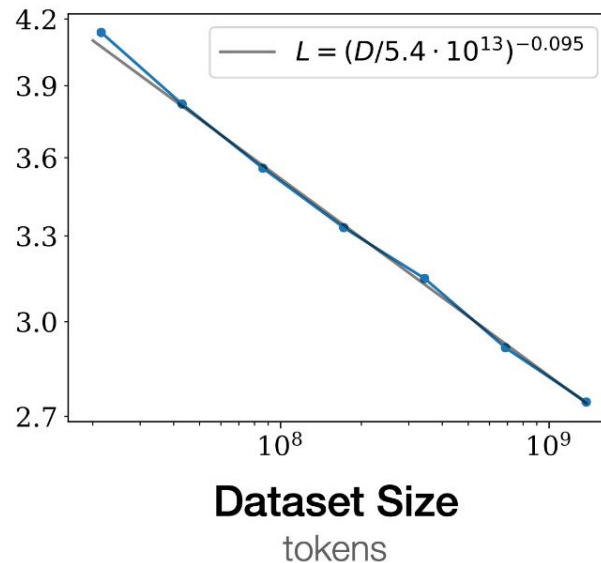
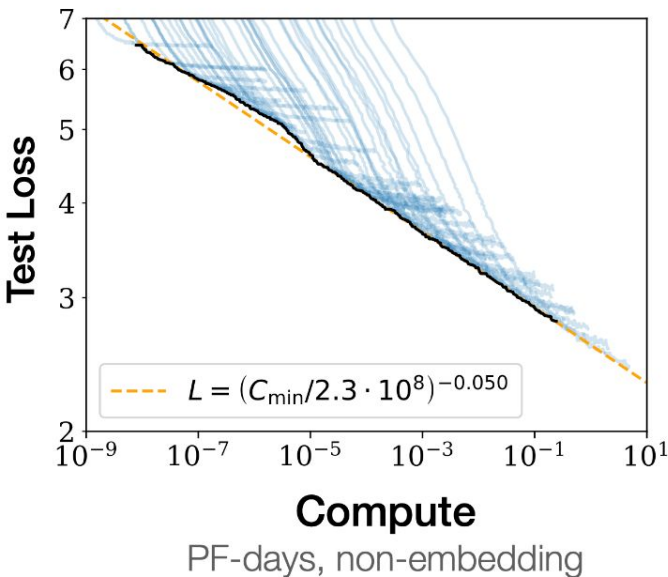


The Compute Allocation Problem

- Hero runs are expensive
- Guessing does not scale
- How big of a model? How long to train?

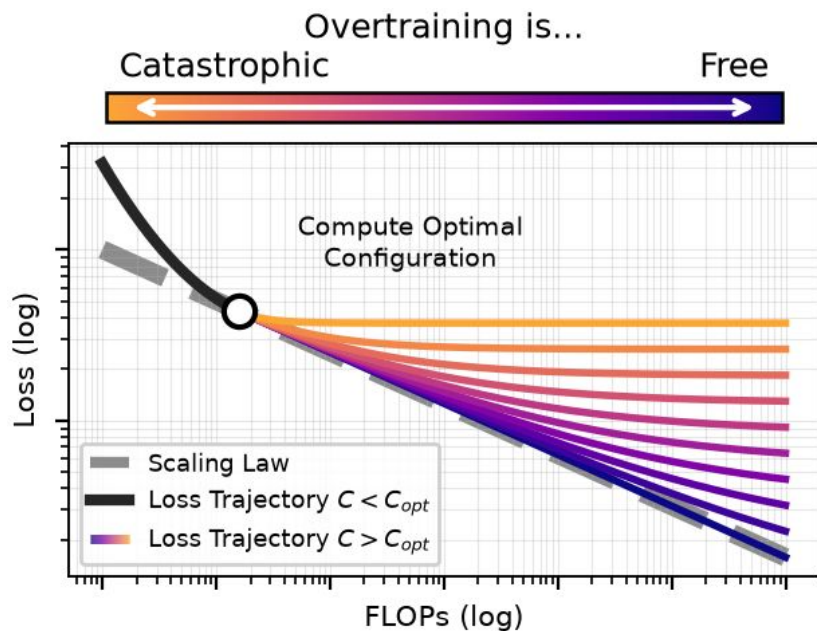
What are Scaling Laws

Hestness / Amodei -> Kaplan -> Hoffmann -> Modern Day



Why Should I Care

- Avoid spending unnecessary compute
- Compute efficiency = less money = longer training on bigger model = better model
- Sometimes we want to overtrain. When is overtraining “safe”?
- Pre-training still matters (apparently)



...and Relatively Little About Diffusion Scaling

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Bigger is not Always Better: Scaling Properties of Latent Diffusion Models

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Abstract

We study the scaling properties of latent diffusion models (LDMs) with an emphasis on their sampling efficiency. While improved network architecture and inference algorithms have shown to effectively boost sampling efficiency of diffusion models, the role of model size—a critical determinant of sampling efficiency—has not been thoroughly examined. Through empirical analysis of established text-to-image diffusion models, we conduct an in-depth investigation into how model size influences sampling efficiency across varying sampling steps. Our findings unveil a surprising trend: when operating under a given inference budget, smaller models frequently outperform their larger equivalents in generating high-quality results. Moreover, we extend our study to demonstrate the generalizability of these findings by applying various diffusion samplers, exploring diverse downstream tasks, evaluating post-distilled models, as well as comparing performance relative to training compute. These findings open up new pathways for the development of LDM scaling strategies which can be employed to enhance generative capabilities within limited inference budgets.

1 Introduction

Latent diffusion models (LDMs) (Rombach et al., 2022), and diffusion models in general, trained on large-scale, high-quality data (Lin et al., 2014; Schuhmann et al., 2022) have emerged as a powerful and robust framework for generating impressive results in a variety of tasks, including image synthesis and editing (Rombach et al., 2022; Poell et al., 2022; Delveccio & Siddharth, 2023; Ren et al., 2023; Qi et al., 2023), video creation (Miri & Patel, 2023; Mei et al., 2023; Wu et al., 2023; Singer et al., 2022), audio production (Liu et al., 2023a), and 3D synthesis (Liu et al., 2023; Liu et al., 2023b). Despite their versatility, the major

^{*}This work was done during an internship at Google.

[†]This work was done while at Google.

SCALING LAWS FOR DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers (DT) have already achieved appealing synthesis and scaling properties in content recreation, e.g., image and video generation. However, scaling laws of DT are less explored, which usually offer precise predictions regarding optimal model size and data requirements given a specific compute budget. Therefore, experiments across a broad range of compute budgets, from 1×10^7 to 6×10^8 FLOPs are conducted to confirm the existence of scaling laws in DT for the first time. Concretely, the loss of pretraining DT also follows a power-law relationship with the involved compute. Based on the scaling law, we can not only determine the optimal model size and required data but also accurately predict the text-to-image generation loss given a model with 1B parameters and a compute budget of 1.5×10^7 FLOPs. Additionally, we also demonstrate that the trend of pretraining loss matches the generation performances (e.g., FID), even across various datasets, which complements the mapping from compute to synthesis quality and thus provides a predictable benchmark that assesses model performance and data quality at a reduced cost.

1 INTRODUCTION

Scaling laws in large language models (LLMs) (Kaplan et al., 2020; Hestness et al., 2017; Henighan et al., 2020; Hoffmann et al., 2022) have been widely observed and validated, suggesting that pre-training performance follows a power-law relationship with the compute C . The actual compute could be roughly calculated as $C = 6ND$, where N is the model size and D is the data quantity. Therefore, determining the scaling law helps us make informed decisions about resource allocation to maximize computational efficiency, namely, figure out the optimal balance between model size and training data (i.e., the optimal model and data scale) given a fixed compute budget. However, scaling laws in diffusion models remain less explored.

The scalability has already been demonstrated in diffusion models, especially for diffusion transformers (DT). Specifically, several prior works (Mei et al., 2024; Li et al., 2024) reveal that larger models always result in better visual quality and improved text-image alignment. However, the scaling property of diffusion transformers is clearly observed but not accurately predicted. Besides, the absence of explicit scaling laws also hinders a comprehensive understanding of how training budget relate to model size, data quantity, and loss. As a result, we cannot determine accordingly the optimal model and data sizes for a given compute budget and accurately predict training loss. Instead, heuristic configuration searches of models and data are required, which are costly and challenging to ensure optimal balance.

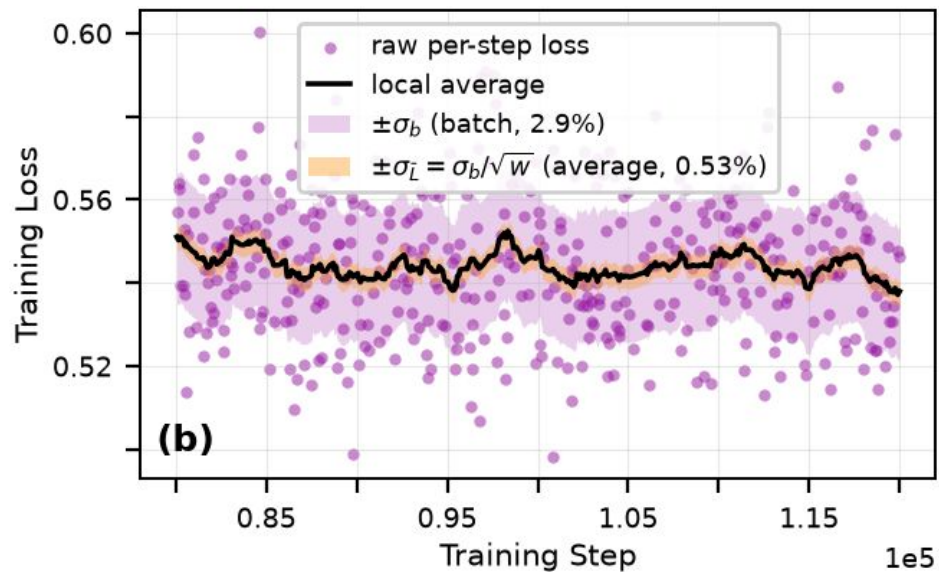
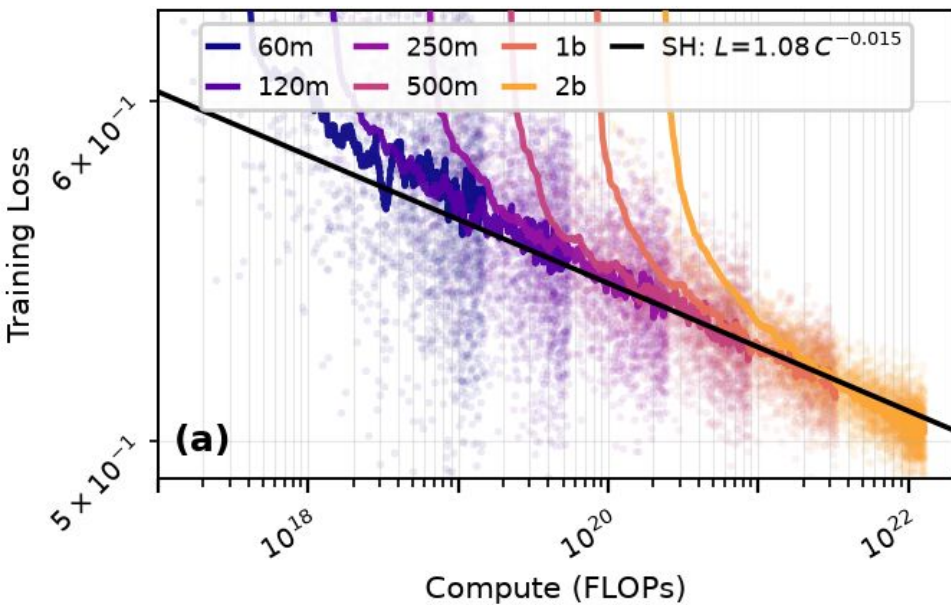
In this work, we characterize the scaling behavior of diffusion models for text-to-image synthesis, resulting in the explicit scaling laws of DT for the first time. To investigate the explicit relationship between pretraining loss and compute, a wide range of compute budgets from 1×10^7 to 6×10^8 FLOPs are used. Models ranging from 1M to 1B are pretrained under given compute budgets. As shown in Fig. 1(a), for each compute budget, we can fit a parabola and extract an optimal point that corresponds to the optimal model size and consumed data under that specific compute constraint. Using these optimal configurations, we derive scaling laws by fitting a power-law relationship between compute budgets, model size, consumed data, and training loss. To evaluate the derived

^{*}Equal Supervision.

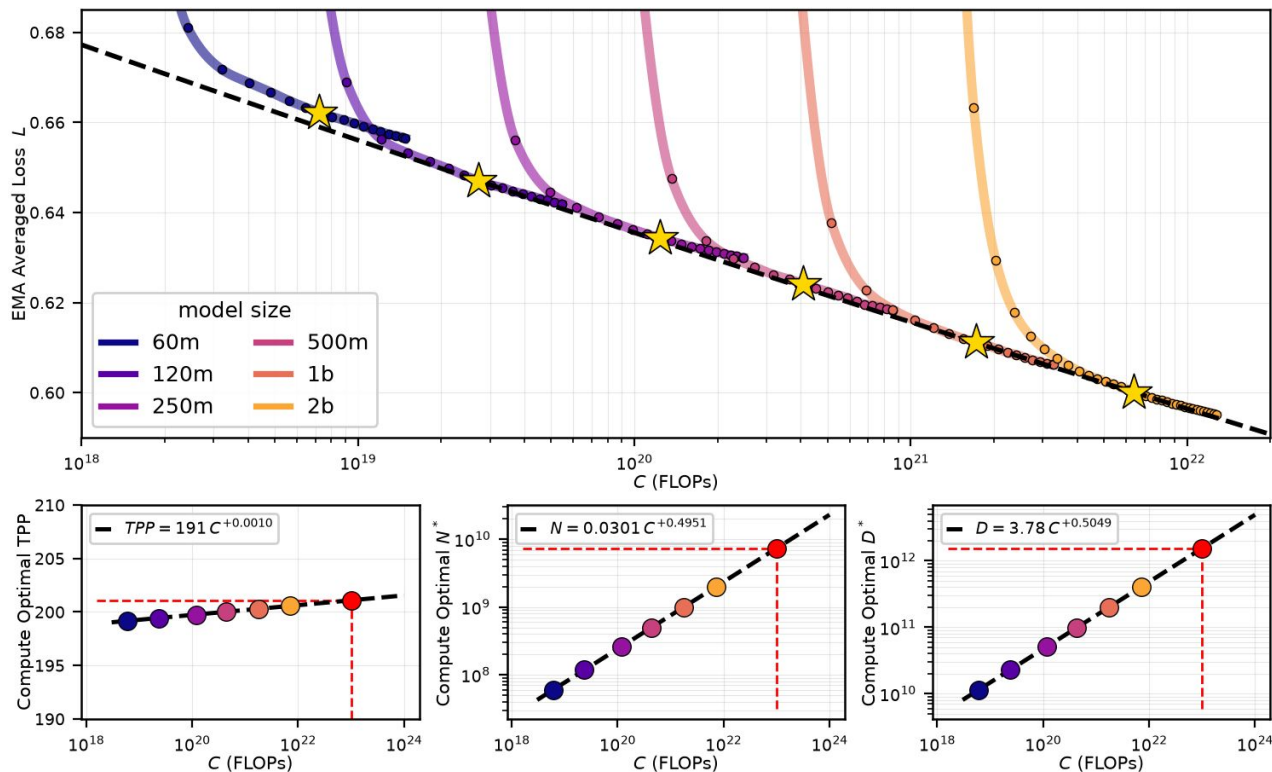
[†]Corresponding Author.

There are good reasons for this

- \$\$\$
- Noise, so much noise
- Community

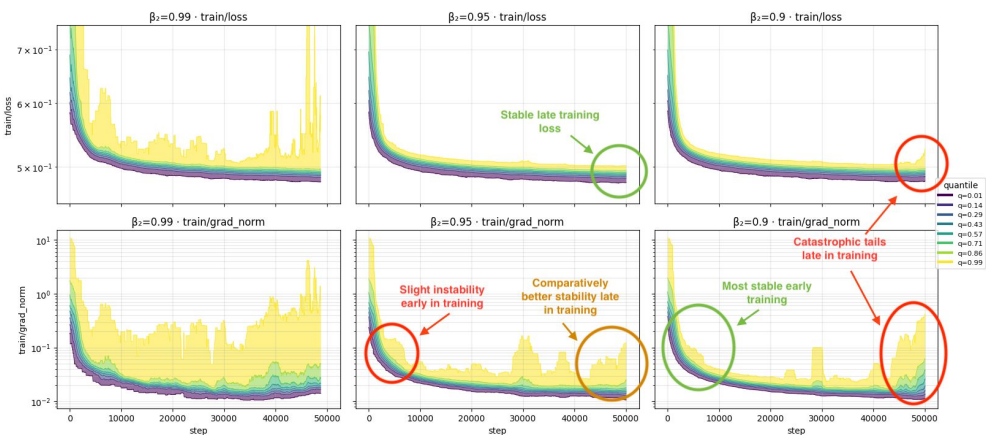
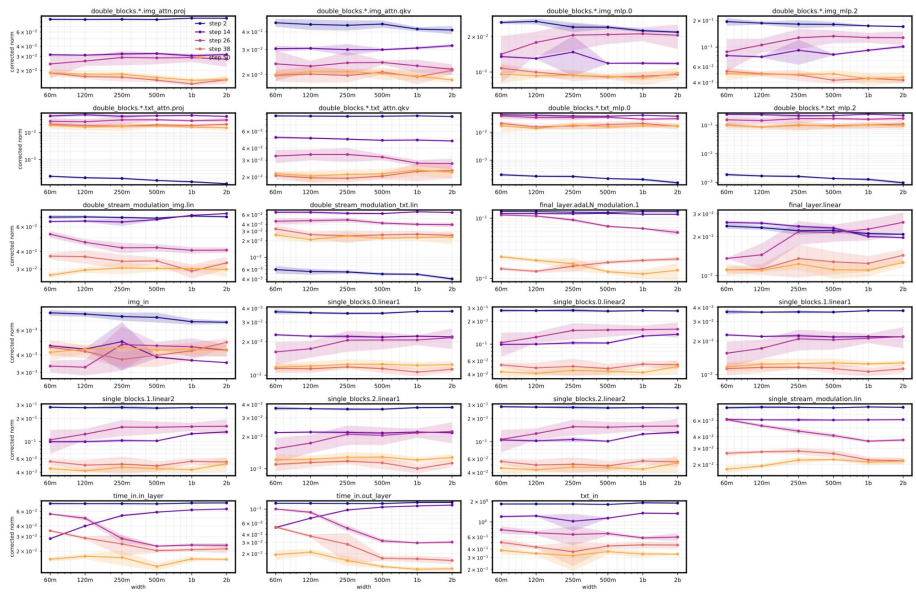
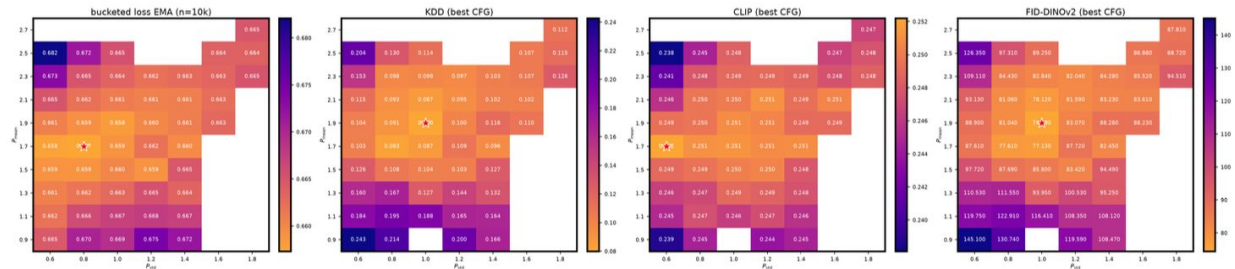


Abra: Scaling for Diffusion

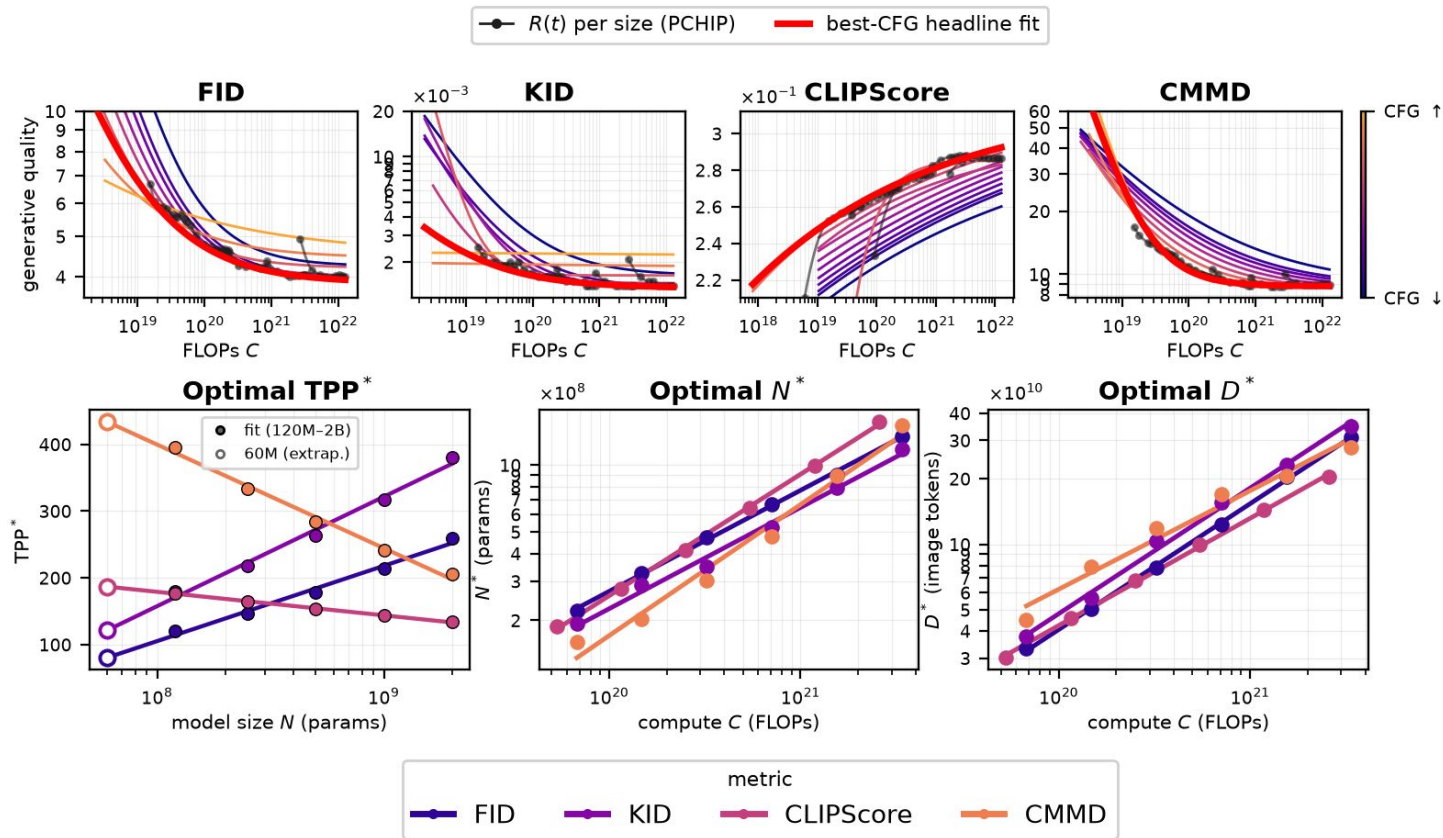


Model Tuning

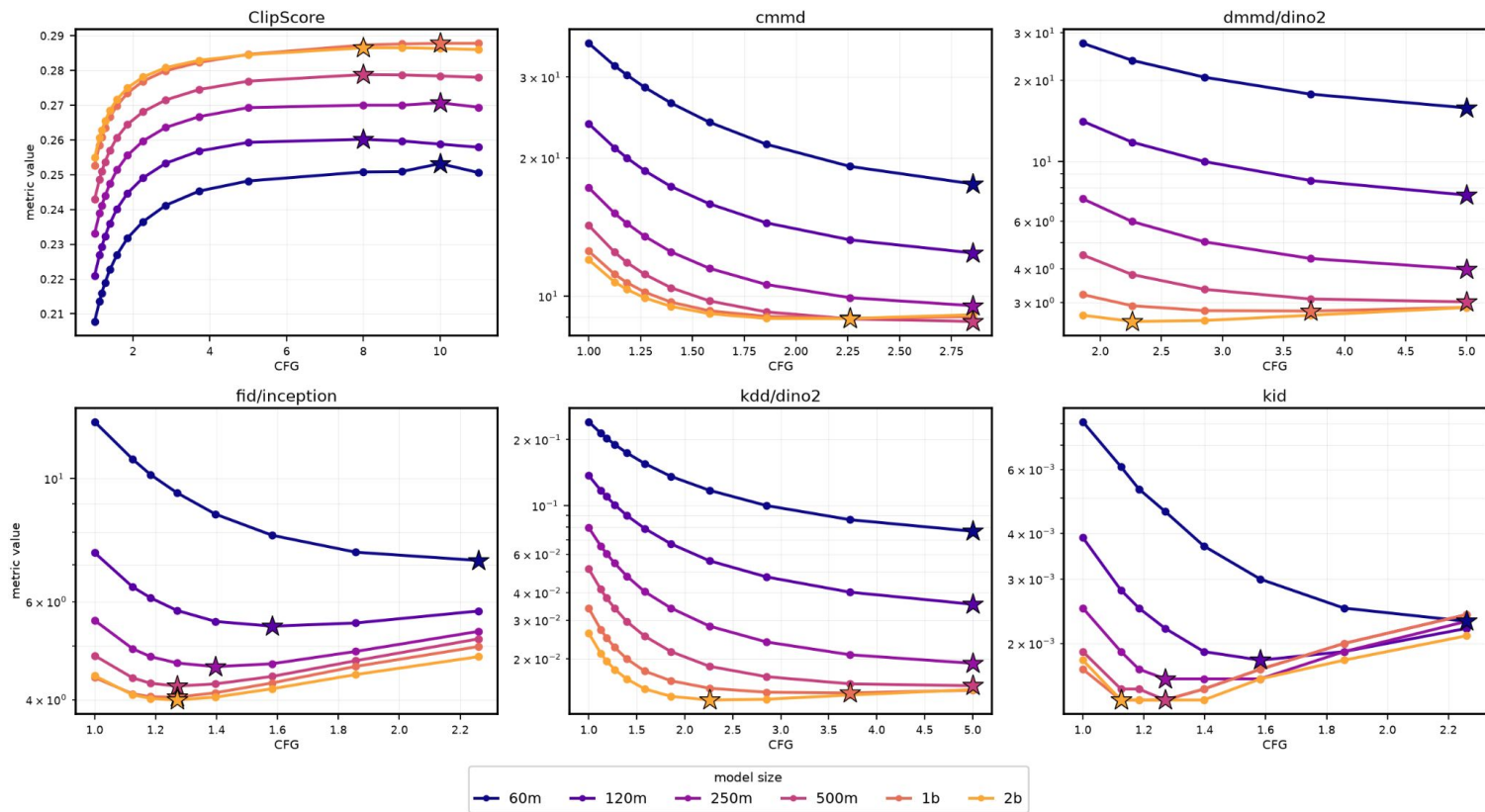
- MuP image
- Beta2 tuning image
- Architectural design: Monotonicity



Generative Metrics Scale



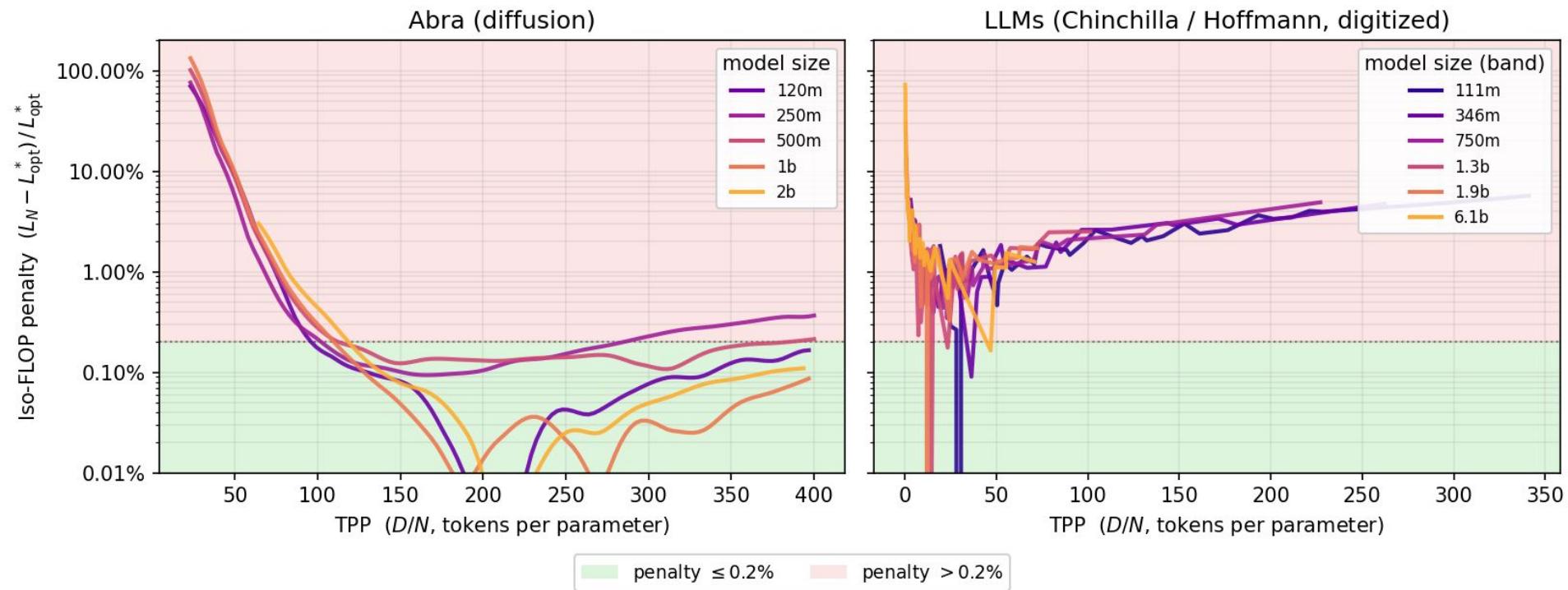
But Optimal CFG is Heavily Metric and Scale Dependent



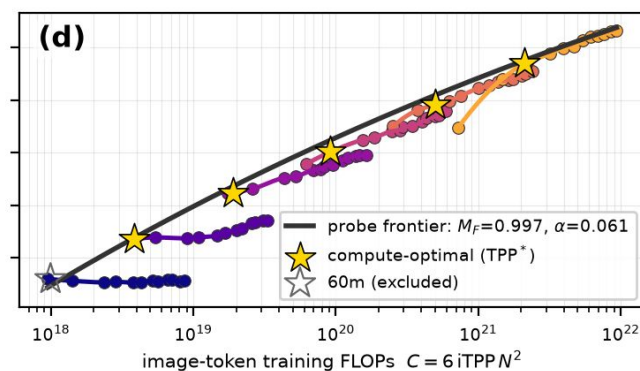
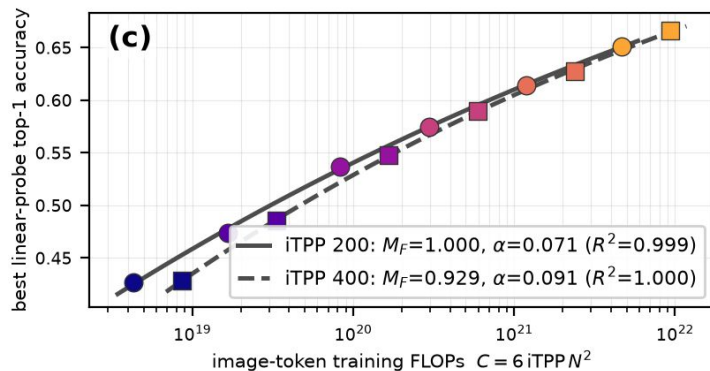
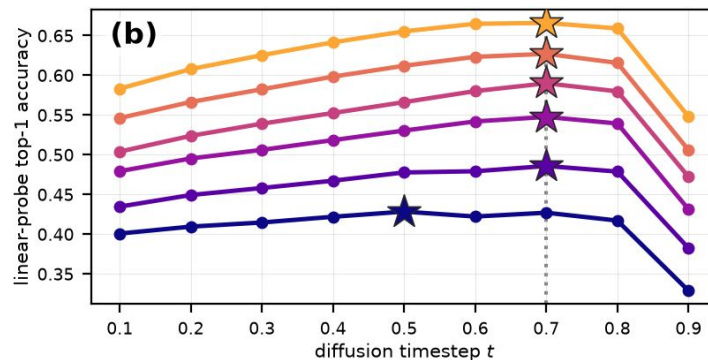
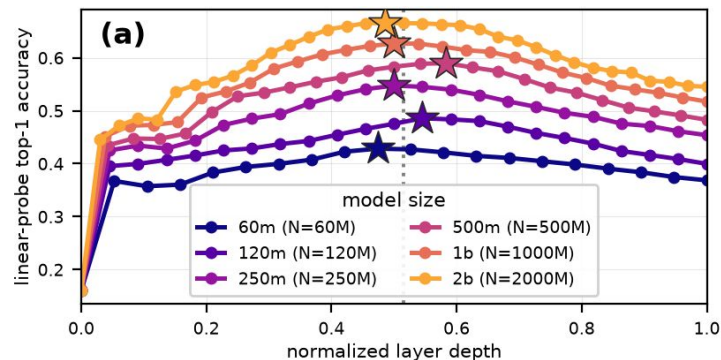
Overtraining is Actually Quite Cheap!

Sub-Optimality Penalty

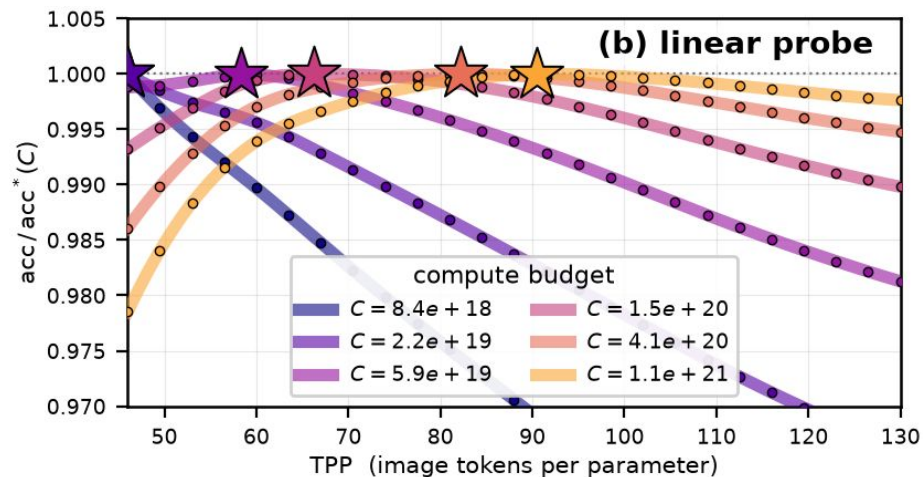
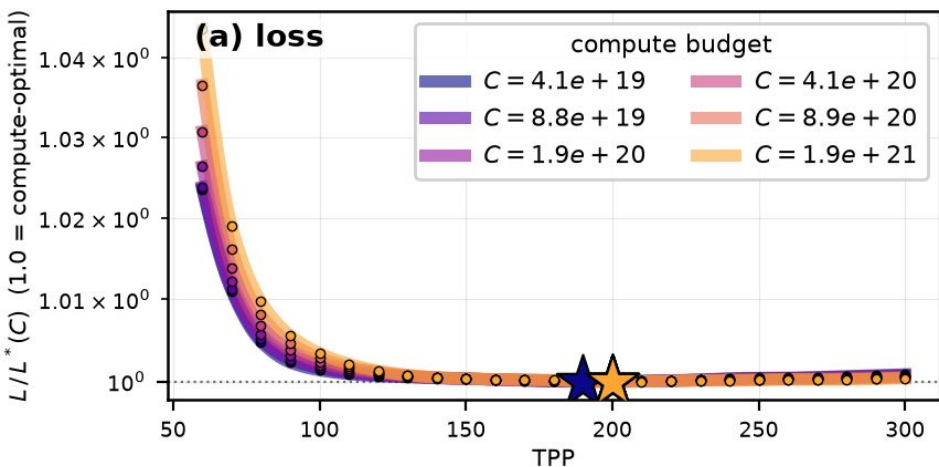
Chinchilla points are figure-digitized (Epoch reconstruction); curves are noisier than real per-run data.



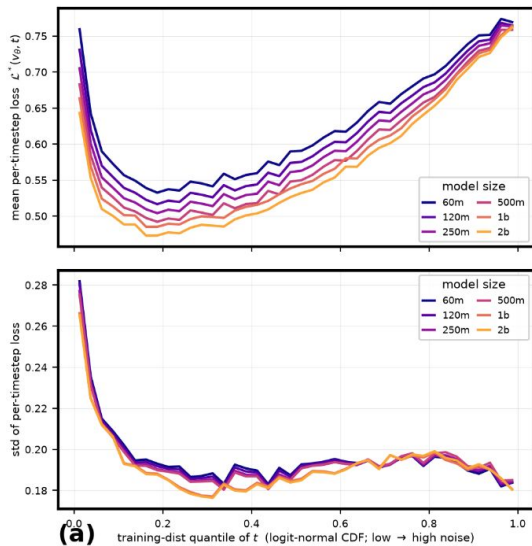
Linear Probing (Visual Understanding) Scales



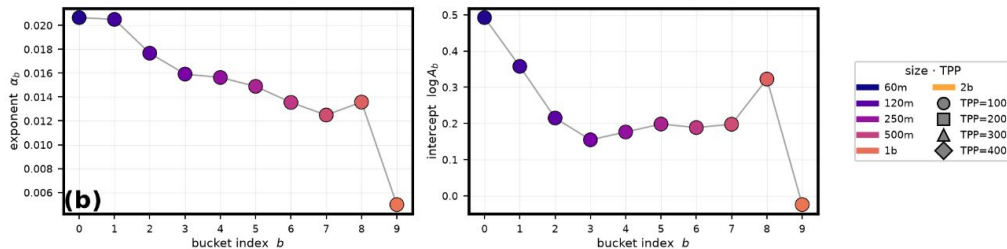
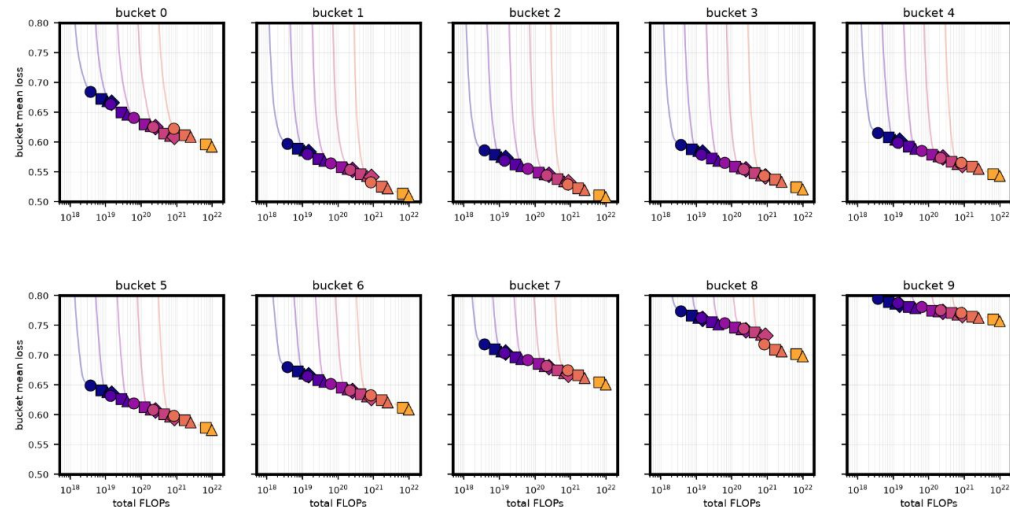
Heterogeneous Scaling Reveals a Dichotomy



Loss Response Curves Scale

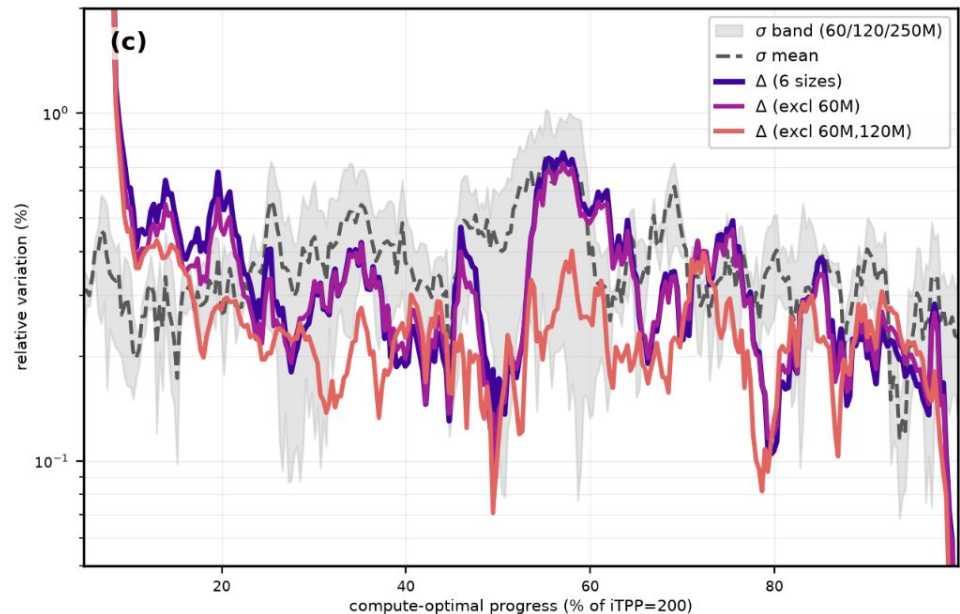
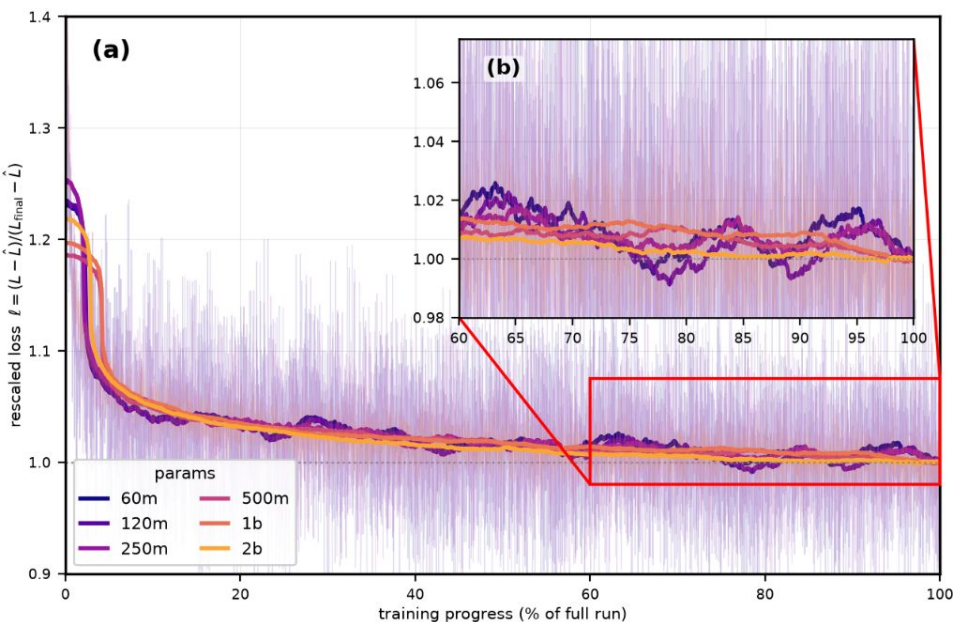


(a)



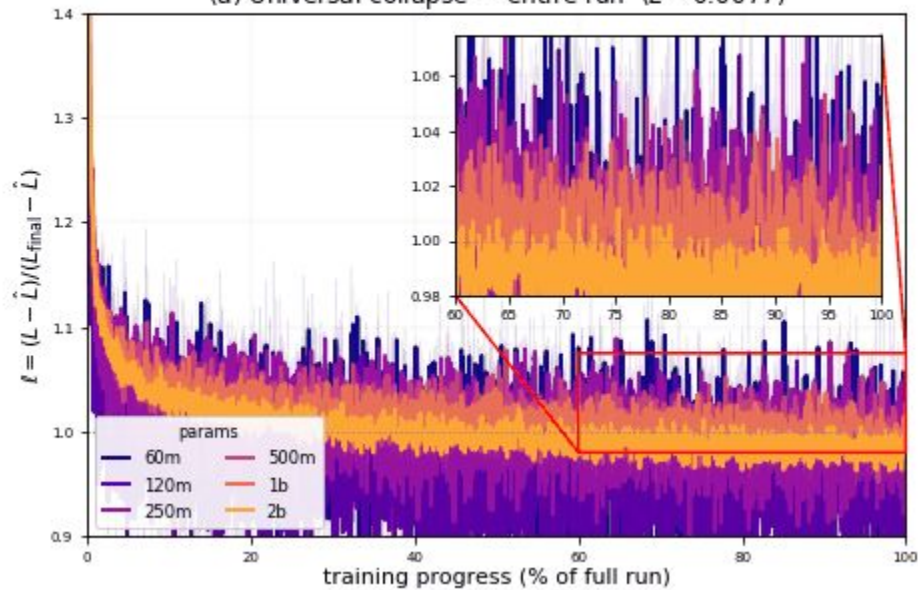
(b)

Scaling Collapse

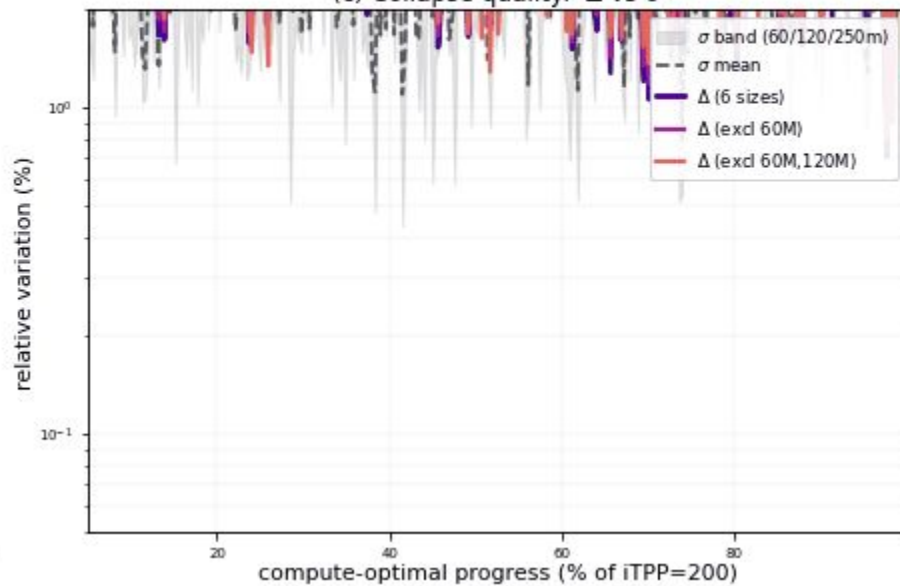


moving-average smoothing window ≈ 100 steps (2 logged points)

(a) Universal collapse — entire run ($\hat{L} = 0.0077$)

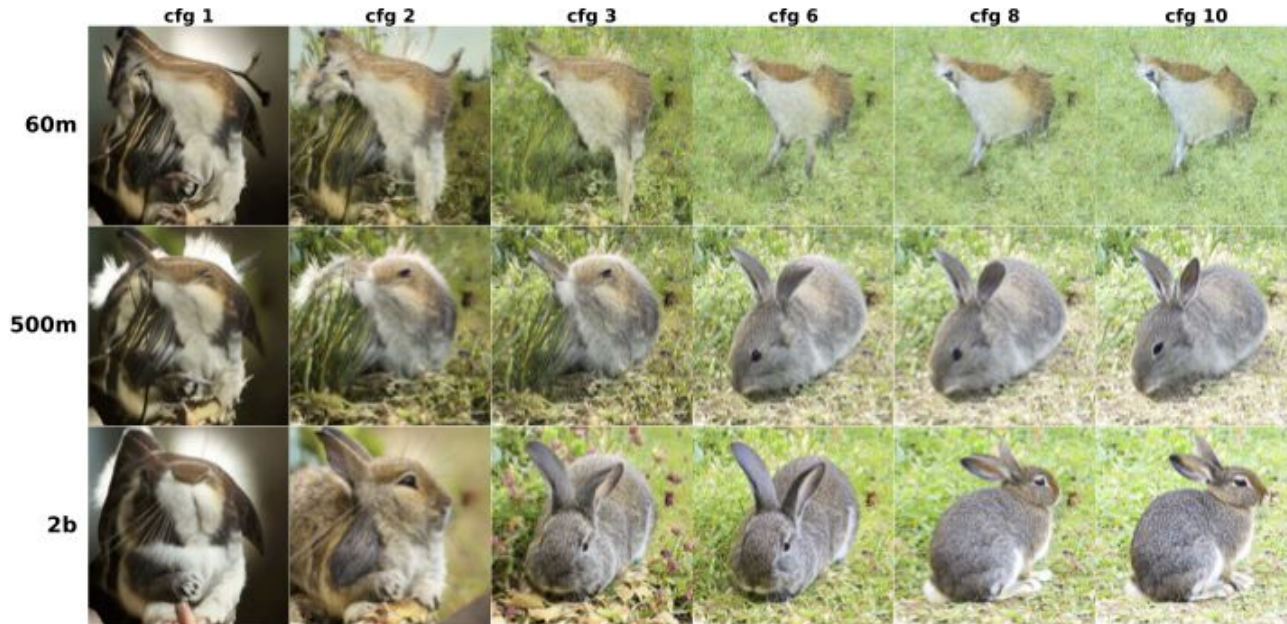


(c) Collapse quality: Δ vs σ



What is a good image?

- Low CFG: Looks great globally, poor fine-grained coherence
- High CFG: Great fine-grained coherence, low global aesthetics



Lessons

1. Diffusion model training is computationally expensive.
2. Overtraining diffusion models is practically free.
3. A few hundred TPP (200-300) is the sweet spot for image diffusion models.
4. Video diffusion models are probably 6-10x more expensive.
5. Tuning CFG is a must when comparing two models.
6. Despite everything in me screaming “NO!”, CLIPScore might be the best quantitative metric.

The Future

- How can we increase the semantic information content per token?
- Running on hardware means we need to make smaller models work.
Distillation?
- Understanding the generative space: CFG / realism / diversity. Low-hanging fruit with very low compute needs.

Scaling Embodied Intelligence

- Just use Astra?
- Bitter lesson strikes again
- Do WAMs beat VLA in the long-run?

Thank you!